

Problem Set 3

AUTHOR
Arihaan Barua

PUBLISHED
February 8, 2026

Question 1

Part (a)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
# A tibble: 8 × 4
  nameFirst nameLast total_HR total_SB
  <chr>      <chr>      <int>    <int>
1 Barry     Bonds        762      514
2 Alex      Rodriguez    696      329
3 Willie    Mays         660      338
4 Andre     Dawson       438      314
5 Carlos    Beltran      435      312
6 Bobby     Bonds        332      461
7 Reggie    Sanders      305      304
8 Steve     Finley       304      320
```

Part (b)

```
# A tibble: 10 × 3
  player      total_W total_S0
  <chr>      <int>    <int>
1 Walter Johnson    417    3509
2 Greg Maddux      355    3371
3 Roger Clemens     354    4672
4 Steve Carlton    329    4136
5 Nolan Ryan       324    5714
6 Don Sutton       324    3574
7 Phil Niekro       318    3342
8 Gaylord Perry     314    3534
9 Tom Seaver        311    3640
10 Randy Johnson    303    4875
```

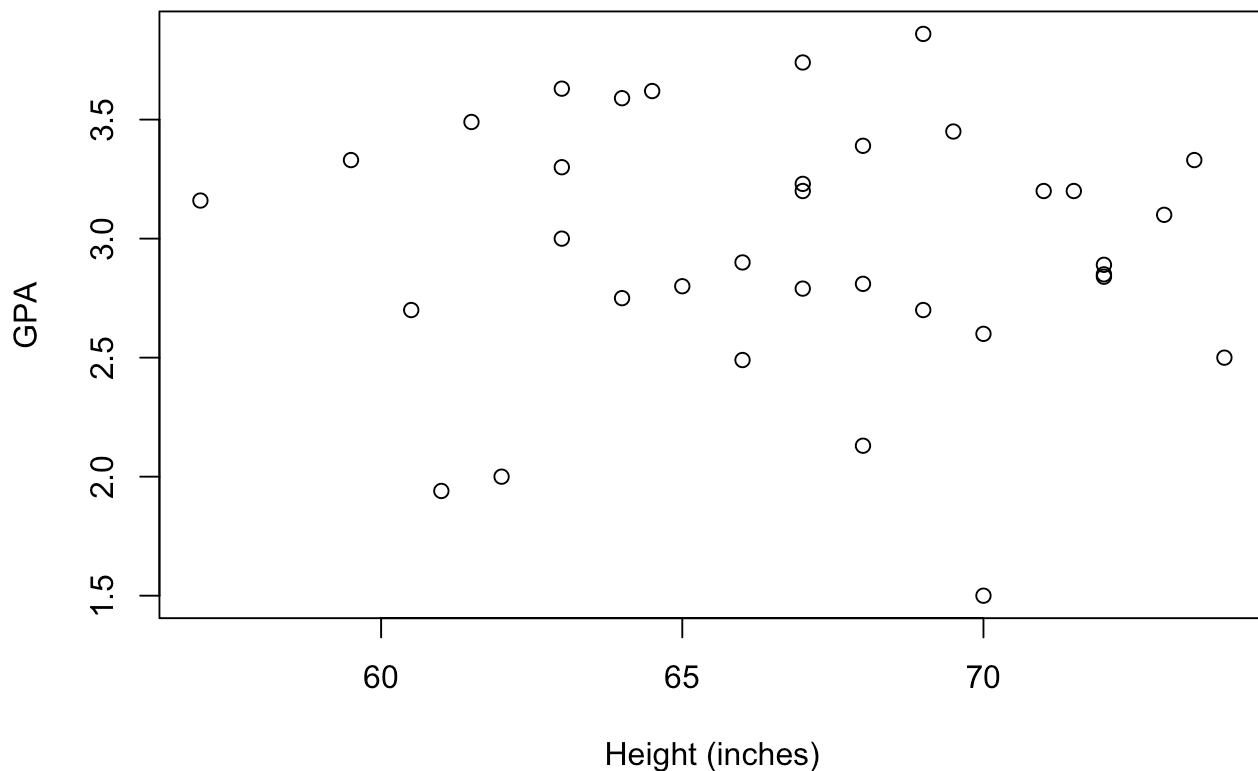
Part (c)

```
# A tibble: 5 × 4
  nameFirst nameLast career_years total_H
  <chr>      <chr>          <dbl>    <int>
1 Pete      Rose             24     4256
2 Ty        Cobb             24     4189
3 Hank      Aaron             23     3771
4 Stan      Musial             23     3630
5 Tris      Speaker           22     3514
```

Question 2

Part (a)

Scatterplot of GPA vs Height



There appears to be a weak positive linear relationship, but the datapoints are pretty evenly spread out.

Part (b)

```
(cor(height_gpa$height, height_gpa$gpa))
```

```
[1] -0.05324126
```

Part (c)

```
model <- lm(gpa ~ height, data = height_gpa)
summary(model)
```

Call:

```
lm(formula = gpa ~ height, data = height_gpa)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.45081	-0.24878	0.00325	0.35622	0.90263

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.410214	1.434616	2.377	0.0234 *
height	-0.006563	0.021428	-0.306	0.7613

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

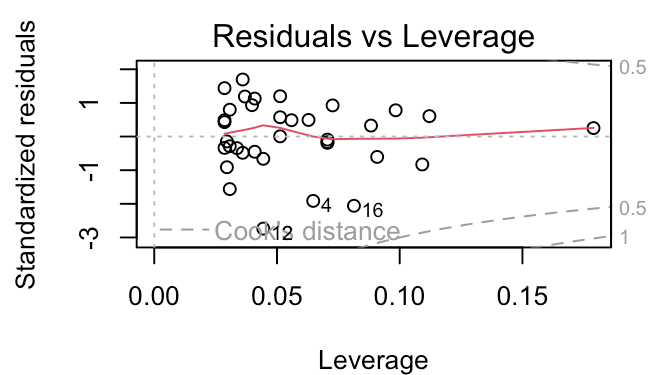
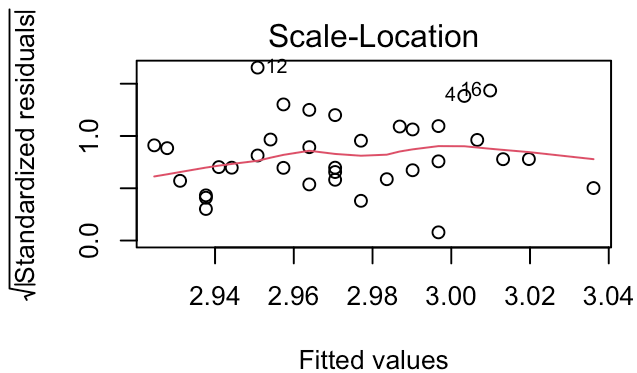
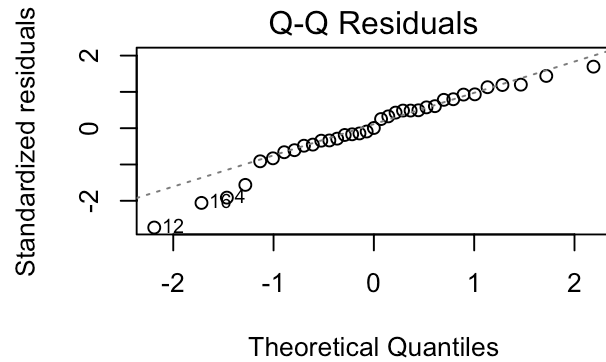
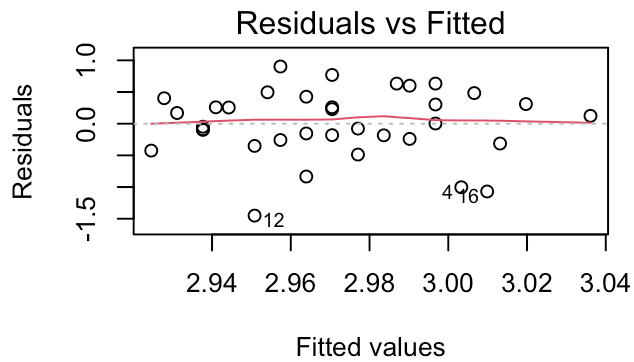
Residual standard error: 0.5423 on 33 degrees of freedom

Multiple R-squared: 0.002835, Adjusted R-squared: -0.02738

F-statistic: 0.09381 on 1 and 33 DF, p-value: 0.7613

The estimated slope represents the change in mean GPA for every one-inch increase in height. In this dataset, the slope is very small, suggesting that height doesn't much effect on GPA.

Part (d)



Part (e)

```
summary(model)$r.squared
```

[1] 0.002834632

The R^2 value is very small, meaning that height explains only a small proportion of the variability in GPA. This means that almost all variation in GPA is due to other factors.

Part (f)

This data set shows that there is little to no meaningful relationship between a student's height and their GPA. Both the correlation coefficient and R^2 value indicate that height is not a useful predictor of academic performance in this sample.